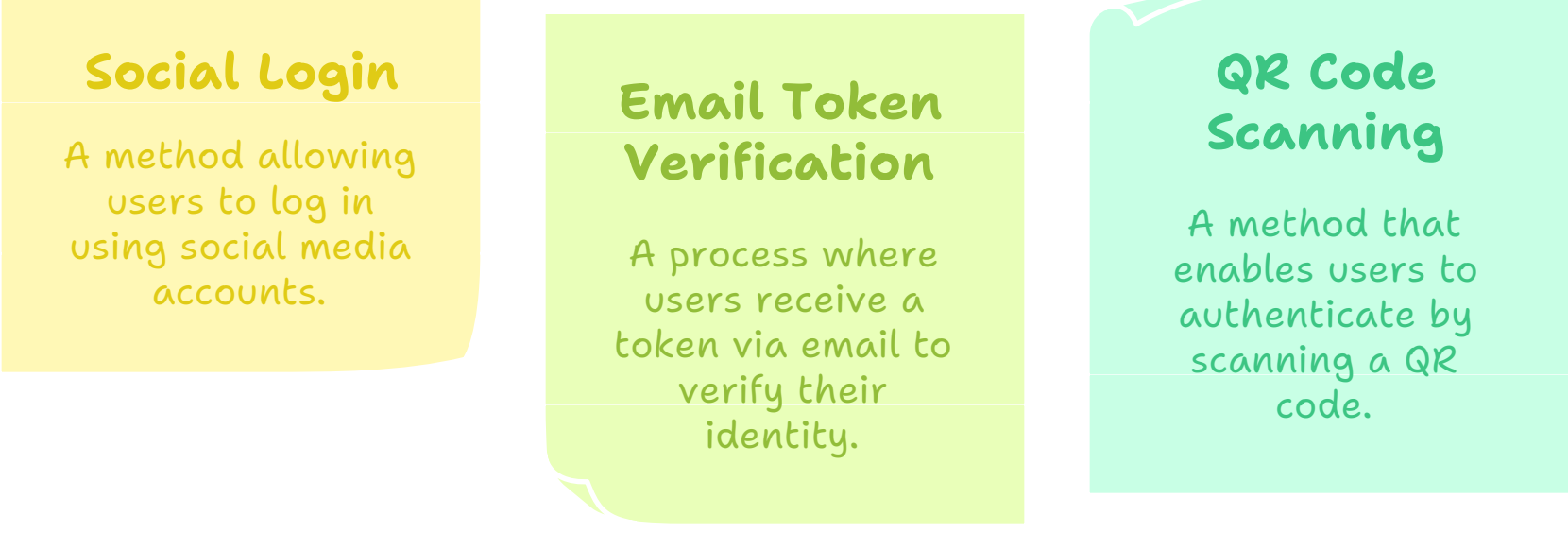


AuthGuard Authentication System

Overview

This document outlines the communication flow between the frontend, backend, and mobile app for the AuthGuard authentication system. The system employs various authentication methods, including social login, email token verification, and QR code scanning, to provide a seamless user experience.

Authentication Methods



Abstract

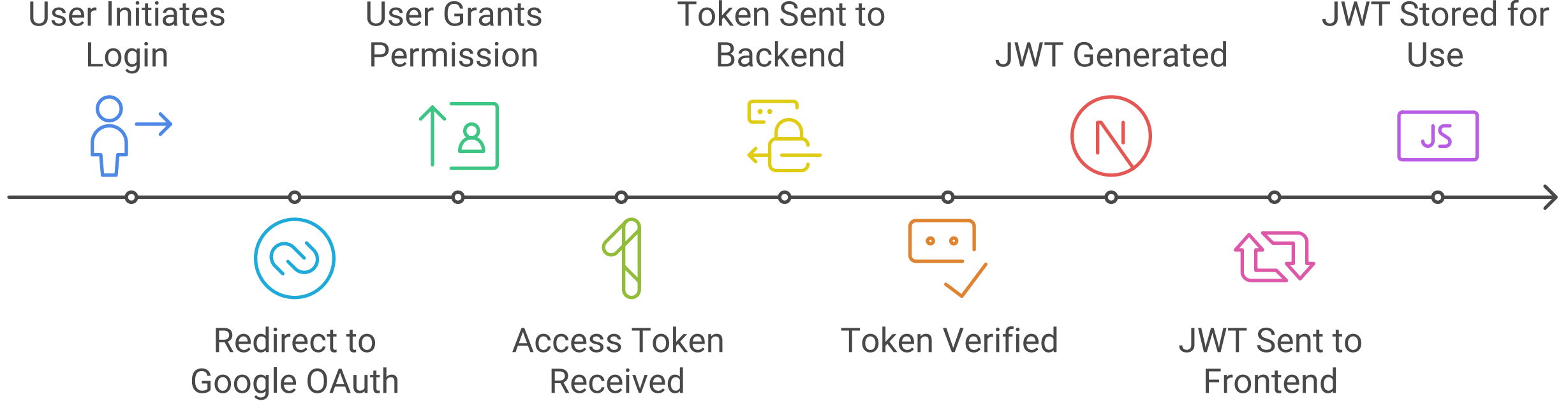
AuthGuard is an innovative authentication application that leverages multiple methods to authenticate users effectively. This document details the interactions between the frontend, backend, and mobile components of the system. It highlights how social login utilizes Google access tokens, the process of sending one-time tokens via email, and the QR code authentication mechanism that allows mobile users to authenticate on the web. Understanding these interactions is crucial for developers and stakeholders involved in the implementation and maintenance of the AuthGuard system.

Communication Flow

1. Social Login

- Frontend:** The user initiates the login process by selecting a social login option (e.g., Google).
- Google OAuth:** The frontend redirects the user to Google's OAuth service, where the user grants permission.
- Access Token:** Upon successful authentication, Google returns an access token to the frontend.
- JWT Token Request:** The frontend sends the Google access token to the backend to request a JSON Web Token [JWT].
- Backend:** The backend verifies the Google access token and, if valid, generates a JWT for the user.
- Response:** The backend sends the JWT back to the frontend, which stores it for subsequent requests.

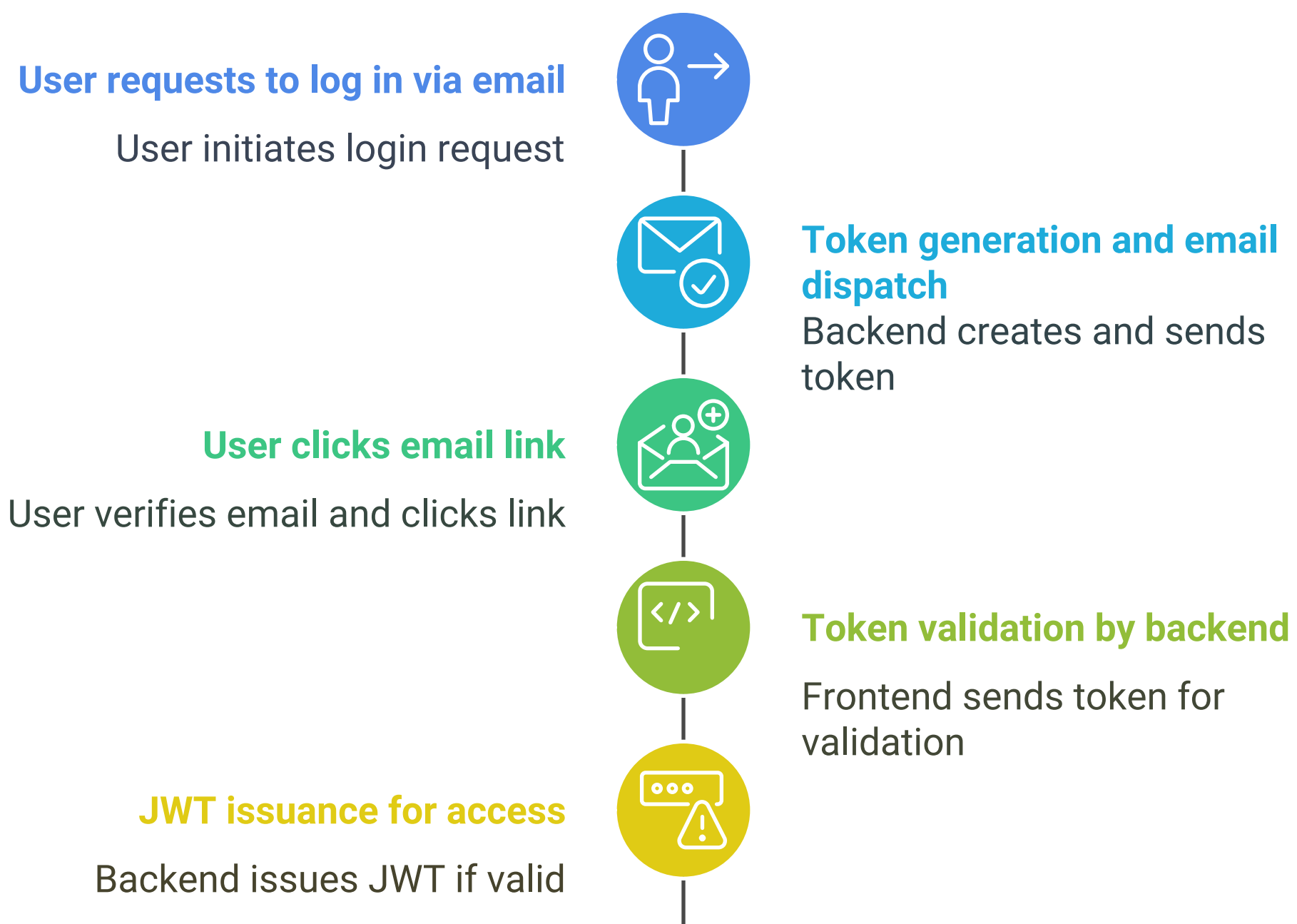
Social Login Authentication Sequence



2. Email Token Authentication

- User Action:** The user requests to log in via email.
- Token Generation:** The backend generates a one-time authentication token and sends it to the user's email address.
- User Verification:** The user checks their email and clicks on the provided link, which contains the token.
- Token Validation:** The frontend sends the token to the backend for validation.
- JWT Issuance:** If the token is valid, the backend issues a JWT to the frontend for authenticated access.

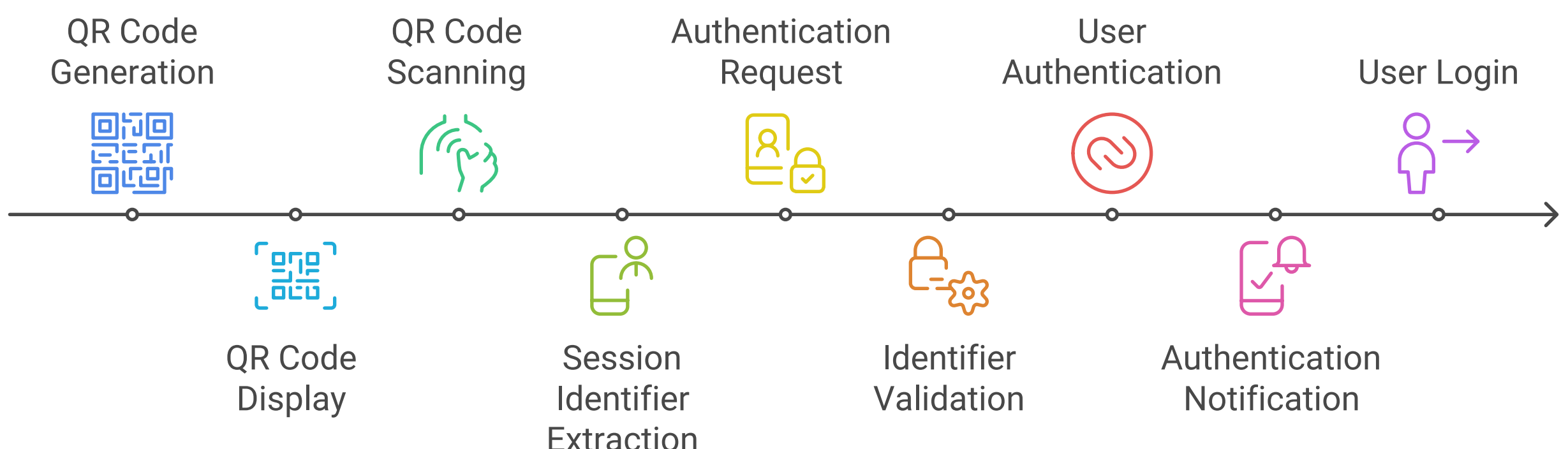
Email Token Authentication Process Timeline



3. Authentication Flow using QR Code and Mobile App

- QR Code Generation (Backend → Frontend)**
 - The **backend** generates a **QR code** containing a unique **session identifier**.
 - The **frontend** displays this QR code to the user.
- QR Code Scanning (Mobile App)**
 - The user opens the **mobile app** and scans the QR code using the camera.
 - The app extracts the **session identifier** from the QR code.
- Mobile Authentication Request (Mobile App → Backend)**
 - The mobile app prompts the user for **confirmation** (e.g., biometric authentication, PIN, or button press).
 - Once confirmed, the **mobile app sends the session identifier** to the backend, signaling approval.
- Backend Verification (Backend)**
 - The **backend validates the session identifier**.
 - If valid, it **authenticates the user on the web session**.
- Web Authentication Confirmation (Backend → Frontend)**
 - The backend notifies the **frontend (web app)** that the authentication was successful.
 - The web app **logs in the user automatically**, completing the authentication process.

QR Code Authentication Process



4. Summary of Communication

- Frontend to Backend:** Requests for JWT tokens, token validation, and session identifiers.
- Backend to Frontend:** JWT tokens, validation responses, and confirmation of mobile authentication.
- Mobile App to Backend:** Authentication requests based on QR code scans.
- Email Service:** Sending one-time tokens to users for email verification.

Conclusion

The AuthGuard authentication system integrates multiple authentication methods to enhance user experience and security. By understanding the communication flow between the frontend, backend, and mobile app, developers can effectively implement and maintain the system, ensuring a robust authentication process for users.